

Chapter: 22

SOLID STATE ELECTRONICS

SOLID STATE ELECTRONICS:

Definition:

“Solid-state electronics is a branch of electronics that deals with the flow of electric current through solid materials, such as semiconductors, insulators, and conductors”.

Explanation:

Solid state electronics involves the design, development, and application of electronic devices and circuits that use these materials to control and manipulate electrical energy.

It has revolutionized the field of electronics, enabling the creation of smaller, faster, and more efficient devices, such as transistors, diodes, integrated circuits, and microprocessors, which are essential components in modern computers, smart phones, and other electronic devices.

The field is constantly evolving, with advancements in materials science, nanotechnology, and quantum mechanics.

TYPES OF SEMICONDUCTORS:

SEMICONDUCTORS:

Definition:

Semiconductors are materials that have a conductivity between conductors and insulators.

Explanation:

In terms of energy bands, semiconductors are the materials which at room temperature have:

1. Partially filled conduction band
2. Partially filled valence band
3. A very narrow forbidden energy gap (of the order of 1 eV) between the conduction and valence bands.

The gap between the conduction band and the valence band is very small; therefore, whenever sufficient energy is provided to the electrons in a semiconductor, electrons jump from the valence band to the conduction band.

In semiconductors the energy gap is low, nearly 0.7 eV. A few electrons can easily transfer from the valence band to the conduction band, which is empty. Therefore, conduction occurs slightly with the rise of temperature as more electrons become excited into the conduction band, so the conductivity increases rapidly with temperature.

Thus, semiconductors are considered as insulators with low energy gaps between the valence band and the next empty band.

Semiconductor like silicon (Si) and germanium (Ge) have resistivity $\sim 10^{-1} \Omega\text{-m}$. The resistivity decreases with temperature. At zero K, the conduction band is empty, and it acts as a perfect insulator.

TYPES OF SEMICONDUCTORS:

There are two types of semiconductors:

1. Intrinsic semiconductors
2. Extrinsic semiconductors.

1. INTRINSIC (PURE) SEMICONDUCTORS:

Definition:

A semi-conductor in its extremely pure form is known as an intrinsic semi-conductor.

Explanation:

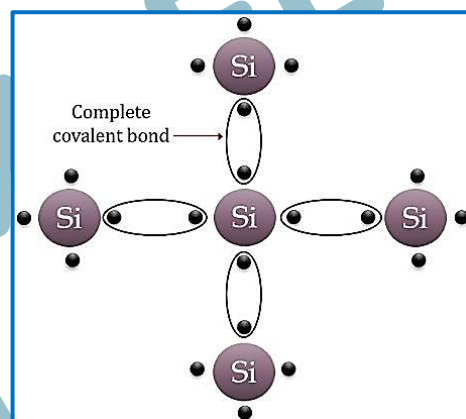
Intrinsic semiconductors are materials that are chemically pure (i.e. they do not contain any significant amount of impurity atoms).

The materials which have intermediate value of resistivity ($10^{-1} \Omega m$), are called intrinsic semiconductors or simply semiconductors.

Intrinsic semiconductors are pure materials with moderate conductivity. These materials have a crystalline structure, and their electrical properties are determined by the arrangement and behavior of their atoms.

Characteristics:

1. These substances belonged to the IV group of the periodic table.
2. They have four valence electrons. Their unit cell is a face centered cube.
3. There are 4 covalent bonds per atom, and each atom is surrounded symmetrically by 4 other atoms forming a tetrahedral crystal.
4. When an electron escapes from a band due to thermal excitation, it leaves a vacant place. This is equivalent to the positive charge of magnitude "e" called "HOLE". The process of entering electron into holes is called "recombination".
5. Pure or intrinsic semiconductors have equal number of electrons and holes.
6. Intrinsic semiconductors have very low conductivity, due to the small number of electrons and holes.



Examples:

The two most used intrinsic semiconductors are silicon (Si) and Germanium (Ge), their forbidden gap is narrow.

2.EXTRINSIC (DOPED) SEMICONDUCTORS:

Definition:

The semiconductors formed after doping are called extrinsic semiconductors.

Explanation:

These are impure semiconductors or doped semiconductors. Doped semiconductors are created by introducing impurity atoms into the intrinsic semiconductor material. This process is known as doping, and it significantly alters the electrical properties of the semiconductor.

Doping produces two types of semiconductors named.

1. N-type semiconductors.
2. P-type semiconductors.

Doped semiconductors are intentionally infused with impurities to enhance their electrical properties, making them essential for modern electronic devices.

DOPING AND PREPARATION OF DOPED SEMICONDUCTOR:

DOPING:

"The process of the addition of impurity in extremely small amount of Penta or trivalent substance to a pure intrinsic semiconductor is called doping."

Extremely small amount means millionth part.

PRODUCTION OF P-TYPE AND N-TYPE SEMICONDUCTORS:

The intentional addition of impurities alters the conductivity and electrical properties of the semiconductor, allowing for the controlled operation of charge carriers.

TYPES OF DOPING:

There are two types of doping:

1. Donor doping.
2. Acceptor doping.

1. DONOR DOPING OR PREPARATION OF N-TYPE SEMICONDUCTOR:

Definition:

"The semiconductor, which is formed by adding extremely small amounts of a pentavalent element (e.g. Arsenic As, Antimony Sb or Phosphorous P etc.) to Germanium or silicon crystal is called N-Type Semiconductor".

Explanation:

In the n-type semiconductor "N" stands for negative semiconductors, the doping is called donor doping whereas the pentavalent impurities are called donor impurities.

It is characterized by an excess of negatively charged carriers, namely electrons, or majority negative charge carriers.

In N-type semiconductors, electrons are the primary charge carriers responsible for the material's electrical conductivity.

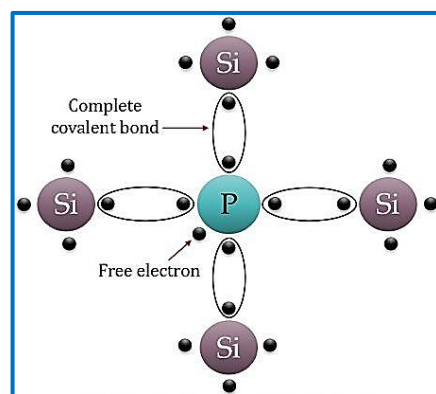
Doping with Donor Impurities:

A small amount of a group-V element (penta-valent) doped into the semiconductor's crystal lattice. Common donor impurities include Boron (B), Phosphorus (P) or Arsenic (As).

These impurity atoms have one more valence electron than the material, creating extra electrons that are free to move and conduct electricity.

Mechanism:

1. Each of the atoms of the doping element forms a bond with four neighbouring atoms. The fifth electron remains surplus. It wanders freely in the lattice.
2. The impurity atom is called donor atom since they donate electrons at the rate of one electron per atom. As shown in the figure.
3. Due to applied electric field free electrons becoming "excited" electrons. They jump to the conduction band from the valence band. Thus, in addition to electrons and holes available, conduction occurs. The n-type of semiconductors are charge carriers.



2. ACCEPTOR DOPING OR PREPARATION OF P-TYPE SEMICONDUCTOR

Definition:

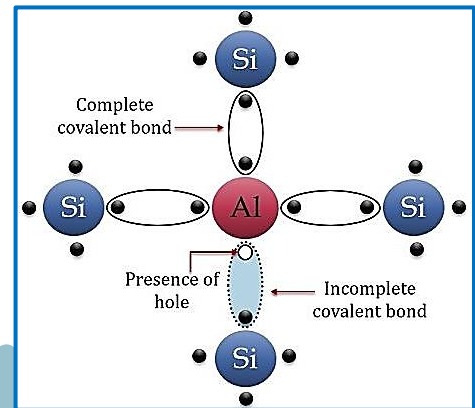
"The semiconductor, which is formed by adding extremely small amounts of a trivalent element (e.g. Boron B, Aluminium Al, Antimony Sb or gallium Ga etc.) to Germanium or silicon crystal is called P-Type Semiconductor".

It is characterized by an excess of positively charged carriers, known as "holes," or majority Positive charge carriers.

Explanation:

In P-type semiconductors, the holes effectively behave as positive charge carriers in the material. These holes are essentially vacancies in the crystal lattice structure, where an electron is missing from a covalent bond.

Semiconductor material, silicon (Si) or Germanium (Ge) is the base semiconductor material.



Doping with Acceptor Impurities:

If element of group-III (tri-valent) is added into the crystal lattice of the semiconductor. Common acceptor impurities include boron (B), Aluminium (Al), Antimony (Sb) or gallium (Ga).

These impurity atoms have one fewer valence electron than the semiconductor material, creating "holes" in the crystal lattice where electrons can move.

Mechanism

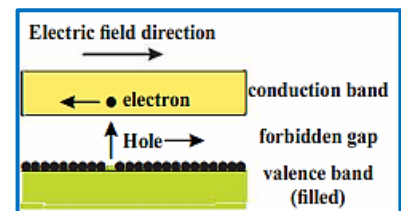
1. Each of the atoms of the doping element forms bonds with four neighbouring atoms. The covalent bond with fourth atom remains incomplete. The vacant place of the electron is called hole.
2. When a nearby electron slips over to fill the hole, it creates another hole at its place. Thus, holes are able to move about in the crystal. The impurities of atoms are called acceptors, since they accept electrons.
3. As the electric field is applied, due to the field the holes are migrated to add conductivity. So, the current is carried through holes. Thus, p-type semiconductors, majority charge carriers are holes. But electrons are a minority charge carrier. In n-type materials electrons are the majority charge carriers.

ELECTRON AND HOLE IN SEMICONDUCTOR:

When an electric field is applied then electric current flow is performed in two ways.

When the temperature increases, then some valence electrons enter into the conduction band and become free electrons. Whenever an electron enters into the conduction band, then a vacant space or a hole is produced in the valence band as shown in figure.

Due to the influence of potential differences or electric field, flow of electrons and holes are produced.



At room temperature, some electrons from the valence band can cross this small energy gap and are accumulated in the conduction band. As a result, an equal number of holes and electrons are produced. This phenomenon is called electron-hole pair Production.

At the temperature above absolute zero, if the electric field is applied then conduction electrons move towards anode and holes move towards the cathode.

So, it can be said that the current flow in the semiconductor is the flow of electrons and holes in the valence and conduction bands in opposite directions.

P-N JUNCTION:

Definition:

The P-N junction is formed when the p-type and n-type semiconductors are joined, is called as P-N junction.

Explanation:

A p-n junction is formed when a crystal of germanium or silicon is grown in such a way that one half is doped with a trivalent impurity and the other half with a pentavalent impurity. Its n-region contains free electrons as majority charge carriers and p-region contains holes as majority charge carriers.

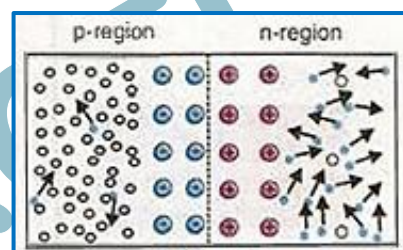
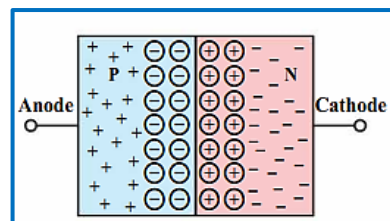
Just after the formation of the junction, the free electrons in the n-region (because of their random motion) diffuse into the p-region.

As a result of this diffusion, a region is formed around the junction in which charge carriers are not present. This region is known as the depletion region.

In this figure, in n-region, dots represent the free electrons, and the small circles show the holes whereas the circles with (+) and (-) signs show the positive and negative ions, which constitute the depletion region.

Due to the charge on these ions a potential difference develops across the depletion region. Its value is 0.7 V in case of silicon and 0.3 V in case of Germanium. This potential difference, called potential barrier, stops further diffusion of electrons into the p-region.

This junction is a crucial component in electronics and forms the basis for many semiconductor devices like diodes and transistors.



DEPLETION LAYER:

Definition:

"The region of a semiconductor in which the density of mobile carrier is too low to neutralize the fix charge density of donors and acceptor, is called as Depletion Layer."

Mechanism:

1. Inside the barrier there is a positive charge on n-type and a negative charge on p-type of the junction; this region is called depletion layer; it consists of no charge carriers. Hence it becomes essential in insulators.
2. The potential barrier gives rise to the electric field, which prevents the respective majority charge carriers from crossing the barrier region.

POTENTIAL BARRIER: NUCLEAR BARRIER

Definition:

"The potential difference that develops across the depletion region of a P-N junction is called Potential Barrier"

Mechanics

1. In a p-n junction, there are holes, i.e. there are majority of charge carriers in p-type and free electrons in n-type semiconductor block.
2. A diffusion of electron and holes take place across the junction; the electrons move across the junction from "n" to "p" type and a positive charge is developed on the "n" side of the junction.
3. At the same time holes move from p-type to n-type, a negative charge is developed on the p-side of the junction.

4. So, a potential difference developed at the junction. It is called a potential barrier or junction barrier. Its value is from 0.1 to 0.7 volt.

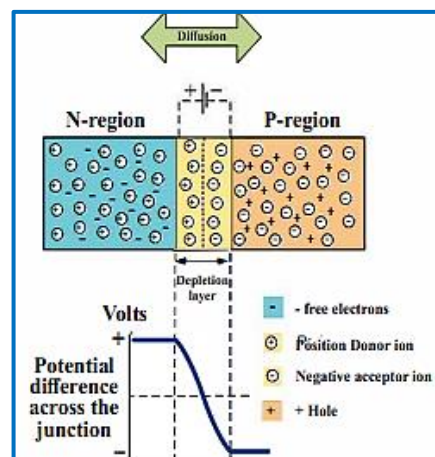
FLOW OF ELECTRONS AND HOLES ACROSS JUNCTION:

Initially, a large density rise exists across the junction who leads to a migration of electrons and holes.

When electrons move from the N-type to the P-type, negative ions are left behind. Simultaneously, holes from P-type migrate toward the N-type, creating positive ions. This process, which is called diffusion, continues until an equilibrium is reached, resulting in a potential barrier zone around the junction, as shown in figure.

The potential barrier, formed due to repulsion between donor and acceptor atoms, creates a depletion layer.

This layer is devoid of free-charge carriers, preventing further movement across the junction.



DIODE:

A diode is a device that has two electrodes.

Definition:

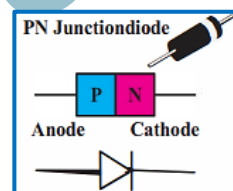
"A diode is a semiconductor device that is formed through P-N junction and used in allowing the flow of electric current in one direction and blocking in the opposite".

The symbol of the P-N junction diode is shown in figure.

PROPERTIES OF DIODE:

Below are some of the common properties of a diode:

1. Diode has the ability to rectify electric current.
2. It can create a potential barrier and make use of its capacitance properties.
3. Diode creates various nonlinear current-voltage characteristics.



FUNCTION OF THE P-N JUNCTION AS DIODE

1. The p-type end is referred to as anode and n-type is referred to as cathode.
2. Anode is represented by an arrow and cathode by a bar or a symbol.
3. A p-n junction consists of regions of low resistivity and a region of high resistivity.
4. It has a property of one directional conduction. It allows the current to flow in one direction and the flow of current is practically zero in opposite direction. Thus, it acts as a diode.

BIASING

DIODE BIASING:

Definition:

"The process in which current is required to flow in desire direction through a semiconductor diode in called Biasing".

TYPES OF BIASING:

There are two types of Biasing:

- (1) Forward biasing and
- (2) Reverse biasing.

1. FORWARD BIAS:**Definition:**

When the P-type is connected to the positive terminal of the battery and the N-type is connected to the negative terminal, the biasing of Diode is called Forward bias.

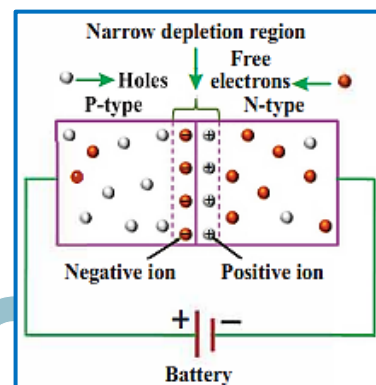
Explanation:

If a p-n junction diode is connected with a source in such a way that its p-side is connected with positive and n-side with negative terminal. It is said to be forward biased.

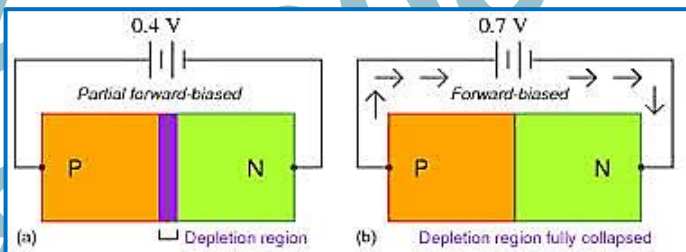
In this condition, the applied electric field and the built-in electric field at the P-N junction are in opposing directions.

Adding both the electric fields gives a resultant electric field, which is found to be smaller than the built-in electric field. So, the depletion region becomes thinner and less resistant.

When the applied voltage is high, the resistance of the depletion region becomes insignificant. At 0.3V to 0.6 V, the resistance of the depletion region in silicon becomes absolutely insignificant, allowing current to flow freely through it.

**Mechanism:**

1. The applied forward potential establishes an electric field. This is opposite to the field due to potential barrier.
2. This electric field reduces the height of the barrier at the junction from 0.3 to 0.1V.
3. When forward voltage is applied this small potential barrier is eliminated. The junction resistance becomes nearly zero.
4. Thus, a low resistance path is established which allow to flow the current in a forward direction.



Simply we can say when a diode is forward biased it allows the current to flow in the circuit.

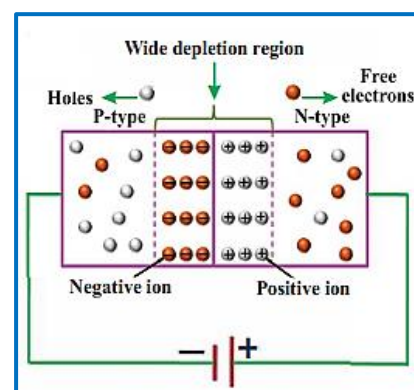
2. REVERSE BIAS:**Definition:**

When the P-type is connected to the negative terminal of the battery and the N type is connected to the positive side, the biasing is called Reverse Bias.

Explanation:

If a p-n junction diode is connected with a source (battery) in such a way that its p-side is connected with negative and n-side with positive terminal of the source. It is said to be reverse biased.

In this condition, the applied electric field and the built-in electric field are both in the same direction. The resultant electric field and the built-in electric field are also in the same direction, resulting in a more resistive, thicker depletion region. If applied voltage increases, it results in thicker and more resistant depletion region.

**Mechanism:**

1. The applied reverse potential establishes an electric barrier. The resultant field becomes stronger; this increases the height of the barrier at the junction.

- The junction resistance becomes high. Thus, a high resistance path is established. Then a very small amount of electric current flows in reverse direction across the junction.

Simply we can say that when a diode is reverse biased it doesn't allow to the current in the circuit.

I-V CHARACTERISTICS OF P-N JUNCTION:

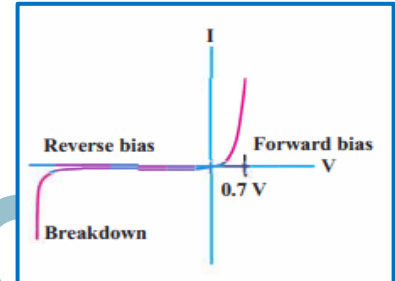
Definition:

The relationship between the voltage across the junction and current through the circuit is known as the (V-I) characteristics of a P-N junction or semiconductor diode.

Explanation:

The V-I characteristics of the P-N junction can be explained in three cases:

- In Zero bias or Unbiased:** At zero bias, no electric current flows through the diode because there's no external voltage applied to enable the movement of electrons or holes.
- In Forward bias:** In forward bias, when the diode voltage (V_d) reaches 0.7 V for silicon and 0.3 V for Germanium, current starts flowing. The current increases gradually at first, creating a non-linear curve until the diode surpasses the potential barrier, after which it operates normally, and the curve steepens linearly with increasing external voltage.
- In Reverse bias:** In reverse bias, only a small leakage current flows, represented to the left of the origin in the graph. This current remains low until the diode breaks down, at which point it can be destroyed unless a high series-resistance limits the current.



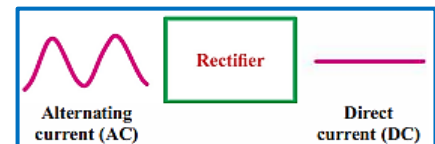
RECTIFICATION:

Definition:

"The process of converting alternating current (A.C) into direct current (D.C) is called rectification".

The device used for this purpose is called rectifier.

Explanation:



Rectification is the process of converting an alternating current (AC) waveform into a direct current (DC) waveform, i.e., creating a new waveform that has only a single polarity.

A diode allows electric current to flow in forward bias condition and blocks the current in reverse bias condition.

Mechanism:

A semiconductor diode can be used as a rectifier. In this ease it is simply connected across an alternating current supply (A.C Voltage Source) with a series resistance. As we know that

- When a diode is forward biased it easily allow to pass electric current i.e. it offers very small resistance i.e. almost zero.
- When it is reverse biased, it doesn't allow the current to flow, it offers large resistance.

Thus, a diode acts as a switch as well as a rectifier.

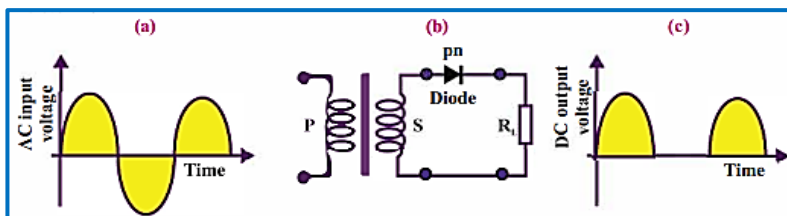
TYPES OF RECTIFICATION:

The rectification is classified into two types according to the output characteristics which are:

- Half-wave rectification and
- Full-wave rectification.

1. HALF WAVE RECTIFICATION:

Since a diode allows AC current to flow only in one direction, it can serve as a rectifier.



As shown in figure (a). The AC source applies a voltage across the diode alternately positive and negative. When the positive cycle of AC voltage passes through the diode, the diode is forward biased and acts as a closed circuit there is current through the resistor R.

During the negative half cycle, the diode is connected with the negative supply which reverses biases the diode, the diode behaves like an open circuit and does not produce the output across the load.

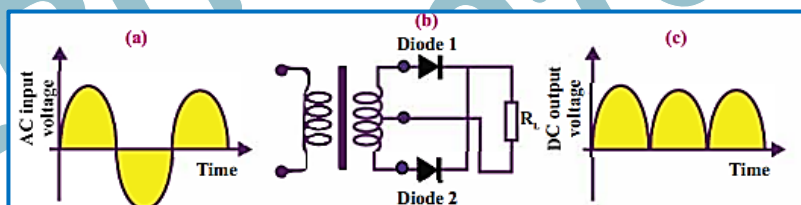
Hence a graph of the voltage V_{ab} across R as a function of time looks like the output voltage shown in figure (b). This is called half wave rectification which seems unidirectional from its output signal.

APPLICATIONS OF HALF-WAVE RECTIFIER:

1. Low power simple battery charger circuit.
2. Fire Alarm circuits.
3. Soldering Iron circuit.
4. Amplitude Modulation (AM) Radio circuits as a Detector.

2. FULL-WAVE RECTIFICATION:

Full-wave rectification uses two diodes (as shown in figure (a, b, c)).



During the positive half cycle of input AC signal, this makes the diode D1 forward biased (acts as closed switch) and diode D2 reverse biased (acts as open switch). Therefore, current flows through the load resistor R.

During the negative half cycle of input AC signal, this makes the diode D2 forward biased and the diode D1 reverse biased. Therefore, the current will flow through diode D2 through, load resistor R and lower half of the secondary winding.

Note that the current through the load is in the same direction for both half cycles of input AC supply. Hence DC output is obtained across the load.

APPLICATIONS OF FULL-WAVE RECTIFICATION:

1. Mobile phones, laptops, charger circuits.
2. Uninterruptible Power Supply (UPS) circuits to convert AC to DC.
3. Our home inverters convert AC to DC.
4. LCD, LED TVs.

APPLICATIONS OF DIODES:

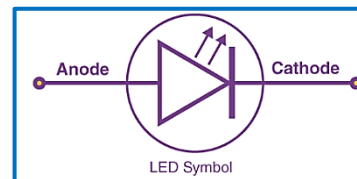
1. LIGHT EMITTING DIODE (LED):

Definition:

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it.

LED Symbol:

The LED symbol is the standard symbol for a diode, with the addition of two small arrows denoting the emission of light.



Construction:

Light emitting diodes (LED) are heavily doped p-n junctions, these are made from special semi-conductors such as gallium arsenide and gallium arsenide phosphide.

Silicon diodes don't work well for this, but compounds like gallium and arsenic create efficient LEDs. Gallium arsenide (GaAs), with a crystal structure similar to silicon, is often used.

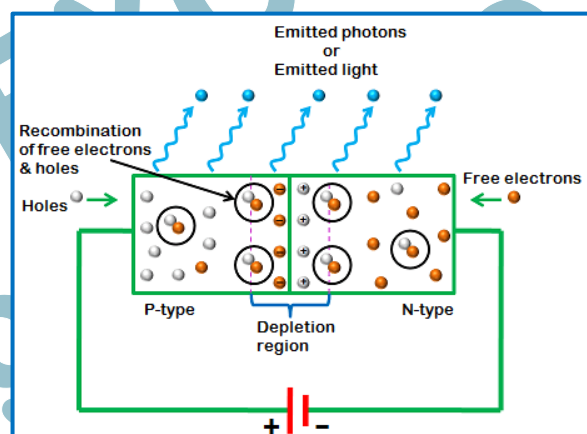
Working:

A light-emitting diode (LED) is a special type of junction in which the potential barrier between p and n sides is such that when an electron combines with a hole during forward bias conduction, a photon of visible light is emitted.

In this process electrons move from the negative side to the positive side, combining with holes, emits a photon with energy similar to the material's band gap.

LEDs allow the current to flow in the forward direction and block the current in the reverse direction.

LEDs, also known as solid-state lighting, last longer (50000 hours vs 2000 hours for regular bulbs), are more efficient, and durable.



Uses:

- These diodes are commonly used as small light sources.
- A specially formed array of seven LED's is used for displaying digits etc., in electronic appliances.
- LED bulbs replace traditional lighting with various applications such as flashlights, streetlights, traffic signals, car brake lights, billboards, and LCD screens.

2. PHOTODIODE:

Definition:

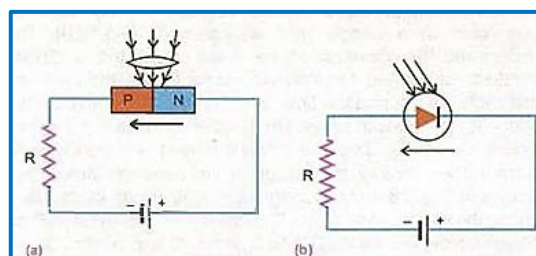
A photodiode is a type of light detector that converts light into current or voltage.

It includes optical filters, built in lenses, and surface areas.

Explanation:

Photo diode is used for the detection of light. It is operated in reverse biased conditions, where a voltage encourages the flow of photocurrent, maximizing sensitivity.

When no light is incident on the junction, the reverse current] is almost negligible but when its p-n junction is exposed to light, the reverse current increases with the intensity of light.



A photo diode can turn its current ON and OFF in nanoseconds. Hence It is one of the fastest photo detection devices.

These detectors are usually made of semiconductor materials like silicon, which can absorb photons of light.

When light particles (photons) hit the semiconductor material, they transfer energy to electrons, creating electron-hole pairs. This process generates an electric current proportional to the incident light's intensity.

In simple terms, brighter light results in a higher current from the photodiode.

USES OF PHOTODIODE:

Photodiodes find widespread use in various applications, including:

1. Optical Communication:

Photodiodes are employed in optical communication systems, such as fiber optics, to detect and convert transmitted light signals into electrical signals.

2. Light Sensors:

They are used in electronic devices like cameras, light meters, and automatic lighting systems to sense ambient light levels.

3. Barcode Readers:

Photodiodes are often used in barcode scanners to detect the reflected light from the barcode.

4. Medical Devices:

In some medical instruments, photodiodes are used for tasks like measuring oxygen levels in blood.

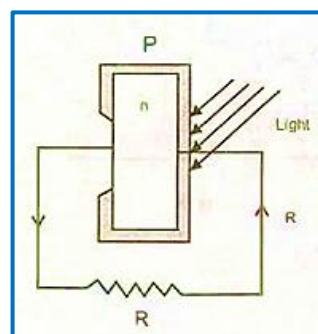
3. PHOTO VOLTAIC CELLS:

Definition:

An extrinsic semiconductor diode, which is used to convert sunlight into electric energy is Photo Voltic Cell or Solar Cell.

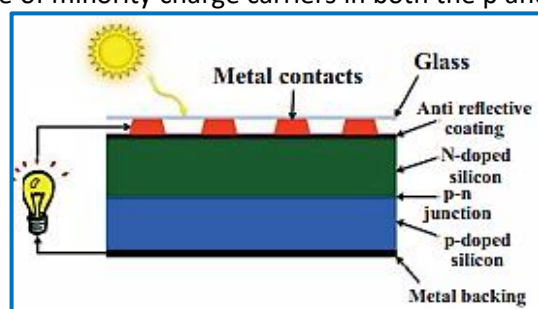
Construction:

- Solar cells or photovoltaic cells are heavily doped P-N junction diodes.
- The basic element of a photovoltaic cell is a semiconductor material, usually made of silicon. Silicon is chosen for its widespread availability.
- Other semiconductor materials, such as cadmium telluride or copper indium gallium selenide, are also used in different types of solar cells.
- It consists of a thick n-type region covered by a thin p-type layer.



Working:

- When a p-n junction has no external bias, and it is exposed to light, it absorbs photons and generate electron-hole pairs. It results in an increase in percentage of minority charge carriers in both the p and n-regions and when they diffuse close to the junction, the electric field due to junction potential barrier sweeps them across the junction. It causes a current flow through the external circuit R. The current is proportional to the intensity of light.
- If the energy of an incident photon is greater than the band gap energy, it creates electron-hole pairs.
- That absorbed photon excites an electron from the valence band and produces a current, when connected to an external circuit, work as source of emf and power.



- A typical silicon P-N junction may produce about 0.6 V. Many are connected in series to produce higher voltage. Such series strings are connected in parallel within a photovoltaic panel i.e., solar panel.
- A good photovoltaic panel can have an output of perhaps 50 W/m², averaged over day and night, sunny and cloudy.

USES OF PHOTODIODES:

1. **Camera:** Light meters, Automatic Shutter Control, Photographic Flash Control
2. **Medical:** CAT Scanners, X-Ray Detection, Pulse Oximeters, Blood Particle Analysis
3. **Automotive:** Headlight Dimmer, Twilight Detectors
4. **Communication:** Fiber Optic Link, Optical Remote Control
5. **Industry:** Bar Code Scanners, Light Pen, Encoders, Surveying Instruments, Copiers Density of Toner.

TRANSISTOR:

Definition:

The three-terminal device formed by sandwiching one type of semiconductor between two other types of semiconductors, is called Semiconductor Transistor."

Explanation:

The word transistor is the combination of two words, first is "transfer" and second one is "resistor".

Transistor means: Transfer of resistor

A transistor is a semiconductor device that amplifies or switches electronic signals by controlling the flow of current between its three layers: base, collector, and emitter.

It acts as a gatekeeper, allowing or blocking the flow of current between the collector and emitter, based on the voltage applied to the base.

It is a very common, active and useful semiconductor device. It has three electrodes.

Construction:

A transistor has three regions NPN or PNP.

1. If the central region is p type which is sandwiched between two n type regions. It is known as n-p-n transistor.
2. If the n type central region is sandwiched between two p type regions. It forms a p-n-p transistor.

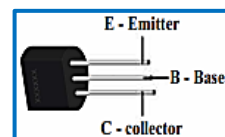
The central region is known as the base and the other two regions are called emitter and collector. Usually the base is very thin, of the order of 10^{-6} m.

The collector is comparatively larger than the emitter. The emitter has greater concentration of impurity, as compared to the collector.

Working:

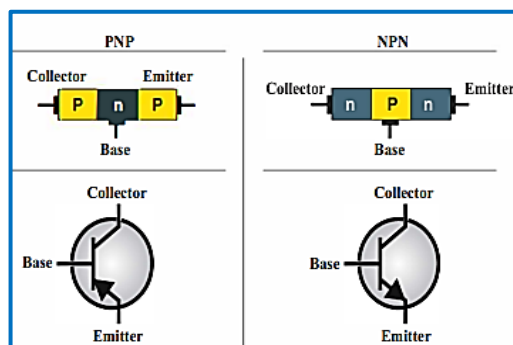
The transistor has three electrodes named as emitter "E", base "B" and collector "C". Were

1. **Emitter:** Supplies free electron or holes.
2. **Base:** Controls the flow of charge.
3. **Collector:** Collects the charges.



Transistor acts as a gatekeeper, allowing or blocking the flow of current between the collector and emitter, based on the voltage applied to the base.

The arrow is always placed on the emitter; it indicates the direction of current flow in normal operation.



NPN TRANSISTOR:

"The NPN transistor consists of two n-type semiconductors that sandwich a p-type semiconductor".

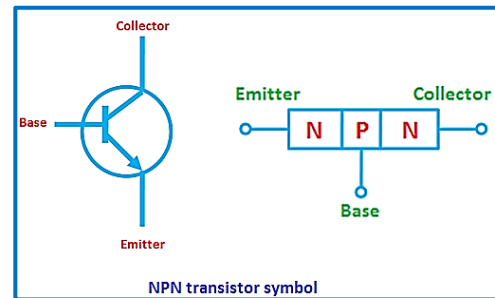
Here, electrons are the majority charge carriers, while holes are the minority charge carriers.

CONSTRUCTION OF NPN TRANSISTOR:

The NPN transistor is made of semiconductor materials like silicon or Germanium.

When a p-type semiconductor material is fused between two n-type semiconductor materials, an NPN transistor is formed.

The NPN transistor features three terminals: emitter, base and collector.



This transistor features two diodes that are connected back-to-back.

1. The diode seen between the emitter-base terminal is referred to as the emitter-base diode.
2. The diode between collector and base terminal is known as collector-base diodes.

The emitter is heavily doped, the base is lightly doped, and the collector is moderately doped.

DIFFERENCE BETWEEN NPN & PNP TRANSISTOR:

No	N-P-N	P-N-P
1	It contains two N-type semiconductors	It contains two P-type semiconductors
2	Its current is directed outwardly.	Its current is directed inwardly.
3	Its current direction is emitter to base.	Its current direction is from base to emitter.
4	It has holes as minority current carriers.	It has electrons as minority carriers.
5	It's switched on when electrons enter the base.	It's switched on when holes enter the base
6	As current reduces at base, transistor does not function across the collector terminal and switches off.	When current is available at base, then transistor switches off.

TRANSISTOR CONFIGURATIONS:

When a transistor is connected in a circuit, one terminal is used as the input terminal, the other terminal is used as the output terminal, and the third terminal is common to the input and output.

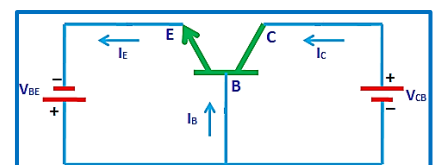
It means that transistor can be connected in the following three configurations. They are:

1. Common base (CB) configuration
2. Common emitter (CE) configuration
3. Common collector (CC) configuration

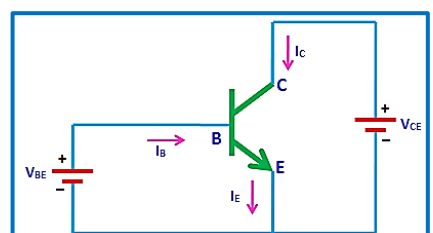
In every configuration, the base-emitter junction J_{BE} is always forward biased, and the collector-base junction J_{CB} is always reverse biased to operate the transistor as a current amplifier.

1. COMMON BASE (CB) CONFIGURATION

In common base configuration, emitter is the input terminal, collector is the output terminal, and base is the common terminal. The base terminal is grounded in the common base configuration. So, the common base configuration is also known as grounded base configuration.

**2. COMMON EMITTER (CE) CONFIGURATION**

In common emitter configuration, base is the input terminal, collector is the output terminal, and emitter is the common terminal. The emitter terminal is grounded in the common emitter configuration. So, the



common emitter configuration is also known as grounded emitter configuration.

3. COMMON COLLECTOR (CC) CONFIGURATION

In common collector configuration, base is the input terminal, emitter is the output terminal, and collector is the common terminal. The collector terminal is grounded in the common collector configuration. So, the common collector configuration is also known as grounded collector configuration.

OPERATION OF COMMON EMITTER NPN TRANSISTOR:

Transistors are categorized as PNP and NPN. For simplicity the NPN transistor is taken as shown in figure.

Here the negative terminal of battery is connected with emitter and positive terminal of battery is connected with collector.

Its action has two requirements:

1. The base-emitter junction is forward biased so that a current I_B is generated.
2. The base-collector junction is reverse biased.

The flow of conventional current is shown in figure.

$$I_E = I_B + I_C$$

As very few electrons flow out of base, therefore I_B is very small as compared to I_C .

It is also found that for a given transistor, ratio of collector current I_C to base current I_B is nearly constant i.e.,

$$\beta = \frac{I_C}{I_B}$$

$$I_C = \beta I_B \dots\dots\dots (1)$$

Where β is the current gain, which is typically 100.

The forward biased caused the electrons to flow from emitter to base. This constitutes emitter current I_E .

In p-type base, only a few electrons less than 5% combine with holes to constitute base current I_B . The remaining 95% crosses over the collector region. This constitutes the collector's current I_C . Thus, the entire collector current flows in the collector circuit.

For completeness, the emitter current is then given by

$$I_E = I_B + I_C$$

By putting the value of I_C , we get.

$$I_E = I_B + \beta I_B$$

$$I_E = (1 + \beta) I_B \dots\dots\dots (2)$$

TRANSISTORS CURRENT EQUATION:

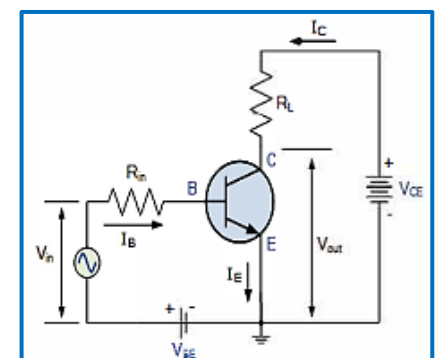
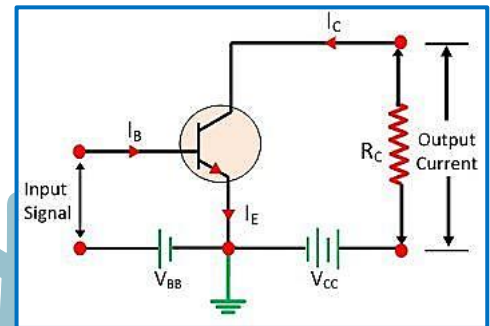
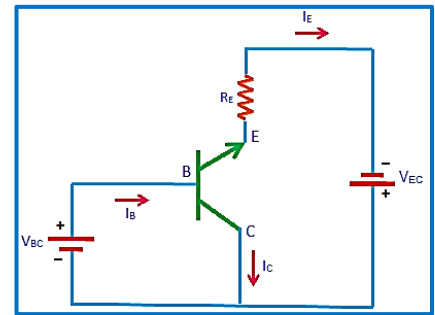
In common emitter Amplifier circuit, the current leaving the transistor must be equal the currents entering it, because the emitter current (I_E) is the sum of the collector current (I_C) and the base current (I_B).

For the common emitter setup:

$$I_E = I_C + I_B$$

Where: " I_C " is the current flowing into the collector terminal, " I_B " is the current flowing into the base terminal and " I_E " is the current flowing out of the emitter terminal.

The common emitter transistor configuration has a substantial current gain because the load resistance (R_L) is connected in series with the collector.



Current Gain:

The current gain is represented by Beta (β), which is the ratio of I_C to I_B for common emitter configuration and the ratio of I_C to I_E for common base configuration is known as Alpha (α), where Alpha is always less than one.

$$\alpha = \frac{I_C}{I_E} \quad \text{And} \quad \beta = \frac{I_C}{I_B} \quad \text{Where } \therefore I_C = \alpha I_E \text{ and } I_C = \beta I_B$$

Due to the physical construction of the transistor, any small change in the base current (I_B) leads to a much larger change in the collector current (I_C). This means that slight variations in the current flowing through the base can effectively control the current in the emitter-collector circuit.

Typically, Beta values range between 20 and 200 for most transistors.

For instance, if a transistor has a Beta value of 100, one electron flows from the base terminal for every 100 electrons flowing between the emitter-collector terminals.

Mathematical Relations between α & β

According to the current equation of common emitter configuration of transistor. i.e.

$$I_E = I_C + I_B$$

Divided both sides by I_C , we get

$$\frac{I_E}{I_C} = 1 + \frac{I_B}{I_C}$$

Where $\frac{1}{\alpha} = \frac{I_C}{I_E}$ and $\frac{1}{\beta} = \frac{I_C}{I_B}$. Therefore, we can write

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta} \dots\dots\dots (1)$$

$$\frac{1}{\alpha} = \frac{1+\beta}{\beta}$$

$$\text{Hence, } \alpha = \frac{\beta}{\beta+1}$$

Similarly, from equation (1), we get.

$$\beta = \frac{\alpha}{1-\alpha}$$

The mathematical relationship between Alpha (α) and Beta (β) expresses the current gain of the transistor.

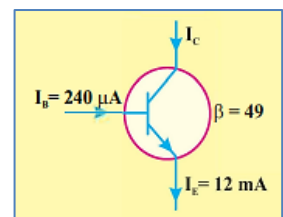
Conclusion:

This type of bipolar transistor common emitter configuration has greater input impedance, current and power gain than that of the common base configuration but its voltage gain is much lower.

The common emitter configuration is an inverting amplifier circuit. This means that the resulting output signal has a 180° phase-shift with regards to the input voltage signal.

WORKED EXAMPLE:

22.1: Find Alpha (α) rating of the transistor shown in diagram. Hence determine the value of I_C using both α and $\beta = 49$ rating of the transistor.

**CHARACTERISTICS CURVES OF THE COMMON EMITTER NPN TRANSISTOR:**

The characteristic curves of a transistor represent the variation of collector current I_C as a function of the collector-emitter voltage V_{CE} for different values of base current I_B .

Transistor characteristic curve is an I-V characteristic of a transistor.

The vertical axis represents the collector current I_C in respect of collector-emitter voltage, V_{CE} . Every change of collector current has something to do with collector-emitter voltage.

The transistor characteristics are divided into four regions:

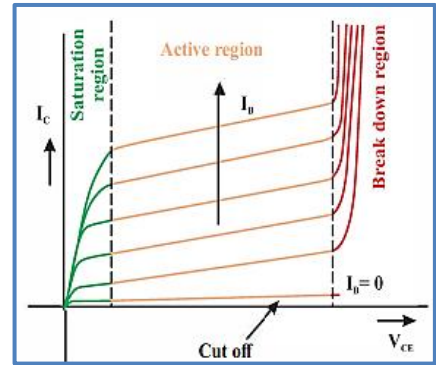
1. Cutoff region
2. Saturation region
3. Active region
4. Breakdown region.

Figure shows the graphs a set of characteristics curves of the transistor.

These graphs show the collector current I_C as function of the collector emitter's voltage V_{CE} , for different values of the base current I_B .

The curve describes the changes in the values of output current I_C against output voltage V_{CE} , keeping the input current I_B constant.

To determine the output characteristics, the base current I_B is held constant for several values. For each fixed value of I_B the voltage between collector and emitter V_{CE} varies in steps and for different V_{CE} the collector current I_C for different value of I_B is called the common emitter output characteristics as shown in the figure.



1. Cutoff Region:

When the transistor-action conditions are not met, the transistor is off and $I_C = 0$. This condition is called cutoff.

Cutoff region is when a transistor acts as an open circuit.

In this region, the transistor remains in the **OFF state**, and no current flows through the collector circuit. Therefore, the cut-off region represents the non-conducting state of the transistor.

Now assume that V_{BE} is forward biased and I_B is above some minimum value. As the collector-emitter junction becomes reverse biased, which we represent here by increasing V_{CE} , transistor action begins to unfold.

2. Saturation region:

In a transistor's characteristic curve, I_C (collector current) initially rises sharply until it reaches a certain point. Beyond this point, I_C continues to increase, but at a much slower rate. This might seem counterintuitive, but the rapid rise is called "**saturation**", indicating full current flow.

The saturation region is where a transistor acts as a short circuit.

The saturation region represents the conducting state of the transistor, similar to a **closed switch**.

3. Active region:

The flat region that follows is when the transistor operates normally, where further increases in base current does not significantly increase collector current I_C . This nearly flat portion of the characteristic curve represents the active region of operation.

Active region is the normal operation of a transistor, where it works as an amplifier.

The transistor operates **normally** in this region and is used for **amplification** purposes.

Thus, the active region is the most important region for practical operation of a transistor as an amplifier.

4. Breakdown Region:

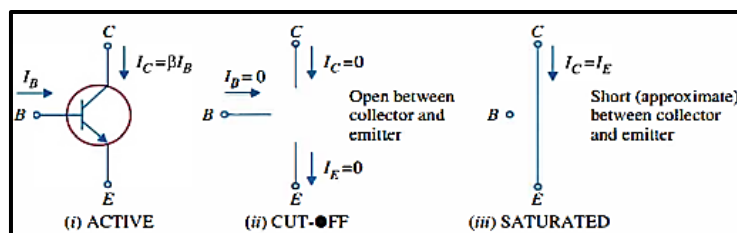
The breakdown region occurs when the collector–emitter voltage V_{CE} exceeds a certain critical value.

In this region, the reverse bias across the collector–base junction becomes very large, causing a sudden and excessive increase in collector current. This phenomenon is known as **breakdown**.

Operation of the transistor in the breakdown region is **undesirable**, as it may damage the transistor due to excessive current and heating.

Conclusion:

Thus, the characteristic curves of a transistor show the cut-off, saturation, and active regions and describe the behavior of collector current with respect to collector–emitter voltage for different base currents.



OPERATION PRINCIPLES, I-V CHARACTERISTICS, BIASING METHODS:

1) OPERATION PRINCIPLES:

1. Emitter, Base, and Collector:

The three layers of a BJT are called the emitter (E), base (B), and collector (C).

The flow of current in an NPN transistor is from the emitter to the collector, and in a PNP transistor, it's from the collector to the emitter.

2. Transistor Action:

The operation of a BJT involves the injection of minority carriers (electrons in NPN or holes in PNP) from the emitter into the base region. This controls the majority carrier flow from the collector to the emitter.

2) I-V CHARACTERISTICS:

Transistors use current-voltage characteristics to show how input voltage relates to output current. This helps understand how transistors amplify or switch signals.

(i) Active Region:

When the transistor amplifies signals, a small current at the base-emitter junction (I_B) causes a larger current from collector to emitter (I_C).

The graph shows a straight line, illustrating signal amplification.

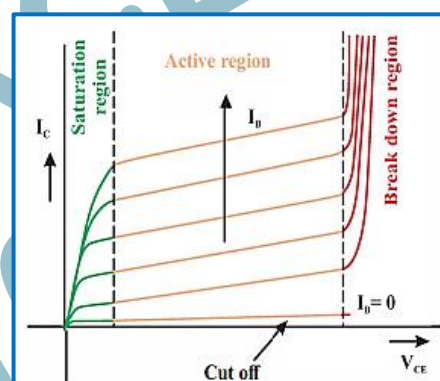
In active region: proper biasing keeps transistors in the active region for linear amplification.

(ii) Saturation and Cutoff:

Beyond the active region, these regions appear.

- **In saturation:** the transistor is ON, allowing maximum collector current.
- **In cutoff:** it's OFF, with minimal collector current.

Understanding these I-V characteristics is essential for designing circuits that utilize transistors for signal processing, amplification, and switching applications.



3) BIASING METHODS:

Biasing is the process of applying external voltages to a transistor to establish a desired operating point for proper amplification or switching.

BIPOLAR JUNCTION TRANSISTOR (BJT) BIASING METHODS:

There are several biasing methods for transistors. Here are some common biasing methods:

1. Fixed Bias (Base Bias):

- Connects a resistor between the base and the power supply to establish a fixed voltage at the base.
- It is simple but not very stable.

2. Emitter Bias (Emitter Stabilized Bias):

- Connect a resistor from the emitter to the power supply, stabilizing the operating point.
- It is more stable than fixed bias.

3. Collector Feedback Bias:

- It combines features of fixed bias and emitter bias for improved stability.
- It uses a resistor network to provide feedback.

4. Voltage Divider Bias:

- It utilizes a resistor divider network to bias the base and provide stability.
- It is more stable than fixed bias but less stable than emitter bias.

TRANSISTORS AS A SWITCH:

A transistor can be used as a switch in electronic circuits, where it functions to either allow or block the flow of current.

The transistor as a switch is used for turning ON or OFF a circuit.

Both types of PNP and NPM transistors are used as switches.

OPERATING REGIONS:

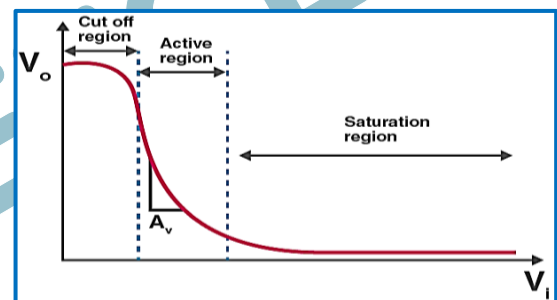
There are two main regions of transistor, working as a switch, these are:

1. Cut-off region.
2. Saturation region.

In the cut-off region, transistor is in turned OFF state and in Saturation region, it is in turned ON state.

1. Transistor can be used as a switch under its cutoff and saturation region. Transistor is in cutoff region when its emitter-base junction is reverse biased.
2. Similarly, it is in saturated region when emitter base junction is forward biased.

These two states act as close and open switch. NPN common emitter configuration is used in this case.



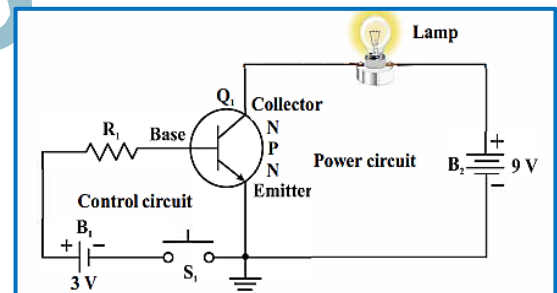
NPN TRANSISTOR AS A SWITCH:

The transistor in figure shows how to control output current with input. The amount of output current is controlled by the forward voltage that controls base current.

CONSTRUCTION:

In figure:

- The input current is base-current for the control circuit.
- The output current is collector-current for the power circuit.
- Q_1 is an NPN transistor, needing a positive V_{BE} for forward voltage.
- The emitter is common to both input control and power circuits.



Common-emitter (CE) circuits are the most common transistor arrangement.

WORKING:

- The base-emitter junction of Q_1 in figure can be forward biased by battery B_1 . Switch S_1 (Virtual) must be closed to apply forward voltage.
- Battery B_2 provides reverse voltage to the collector of Q_1 . Reverse polarity means the collector is more positive than the base.

1. Off State: Cut-off region.

When there is no voltage (or a small voltage) applied to the base, the transistor remains in the OFF state, meaning no current flow between the collector and the emitter.

If emitter base junction is reverse biased, I_B is zero. Therefore $I_C = 0$. In this condition transistor behave as "OFF" switch. If $V_{BE} < 0.7 \text{ V}$, then $I_B = 0 \mu\text{A}$.

As $I_C = \beta I_B$, then $I_C = \beta (0 \mu\text{A}) = 0$

Normally, a transistor allows no output current unless we apply forward voltage to its base emitter circuit.

When switch S_1 is open:

- No current flows in the base-emitter or control circuit because no forward voltage is applied.
- Resistance from emitter to collector of the transistor is very high.
- No current flows in the power circuit, and the lamp doesn't light up.

2. ON State: Saturation region.

When a sufficient voltage is applied to the base, it allows current to flow from the collector to the emitter, effectively turning the transistor ON.

When emitter-base junction is forward biased, there is an increase in " I_B " which cause increase in " I_C " and at maximum value of I_C transistor behave as "ON" switch.

In this case voltage between "C" and "E" is 0.1V.

When switch S_1 is closed:

- A small current flows in the control circuit.
- R_1 limits current in the base circuit.
- Resistance from emitter to collector of Q_1 decreases.
- A large current flows in the power circuit, lighting up the lamp.

Opening switch S_1 , in the control circuit turns off the lamp in the power circuit because resistance from emitter to collector of Q_1 increases again, almost to infinity.

TRANSISTOR AS AN AMPLIFIER:

Definition:

"An amplifier is a device that increases the strength of a weak signal (like sound, radio signal, voltage, current)".

Explanation:

One of the key characteristics of a transistor is that it can be used as an amplifier. Transistors can act as amplifiers while they are functioning in the active region or when it is correctly biased.

The need for a transistor as an amplifier arises when we want to increase or amplify the input signal.

A transistor can take in a very small weak signal through the base junction and release the amplified signal through the collector.

CONSTRUCTION:

For a transistor to work as an amplifier, we usually use the common-emitter configuration. The input is connected in forward-biased, and the output is connected in reverse-biased.

Consider an NPN bipolar junction transistor (BJT) biased to operate in the active region, where the BE junction is forward biased, and the BC junction is reverse biased.

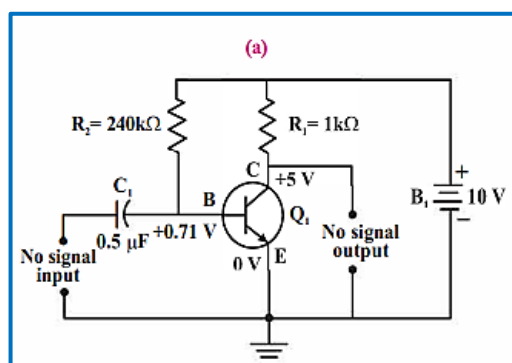
The input signal is applied on the base-emitter junction, and the output is taken through the load in the emitter-collector junction.

WORKING:

Step: 1. Before applying input Signal.

Consider the circuit shown in figure (a), the DC operating voltages are shown in figure (a), without any input signal.

The DC biasing levels are set by the feedback resistor R_2 . The NPN transistor is biased so that the collector to emitter voltage V_{CE} is half of the supply voltage.



Therefore, for the supply voltage of 10 V, the collector voltage is set at one half the total, or +5 V.

The 0.7 V at the base is partially turning on the transistor.

The transistor acts as an amplifier when in this partially turned-on condition. It is the amount of DC forward bias that determines the operating level of the transistor.

Step: 2. After applying input Signal.

An input signal has been added to the amplifier in figure (b), Input is coupled to the base by C_1 . Amplified output is taken from the collector.

The input signal V_{in} is 0.02 V as measured on an oscilloscope.

The measured output signal voltage is 3 V.

The ac gain of the amplifier, therefore, is calculated as

$$\begin{aligned} A_v &= V_{out} / V_{in} \\ &= 3 / 0.02 \\ &= 150 \end{aligned}$$

The output signal (3 V) is 150 times greater than the input signal of 0.02 V. This amplifier stage is said to have a voltage gain of 150.

Uses:

Transistor amplifiers are used:

1. Frequently in RF (radio frequency)
2. OFC (optic fibre communication)
3. Audio amplification, etc.

WORKED EXAMPLE:

22.2: In a common base connection, $I_C = 0.95$ mA and $I_B = 0.05$ mA. Find the value of α .

(2024)

TRANSISTOR CIRCUITS:

COMMON COLLECTOR (CC) CIRCUIT:

In **Common Collector** configuration, the input signal is applied to the base, and the output signal is taken from the emitter, with the collector terminal serving as the common connection. Figure shows this setup.

Here, when the base-emitter junction is forward biased, a small base current I_B causes a much larger collector current I_C to flow.

The emitter current I_E is approximately equal to I_C .

$$\text{Since } I_E = I_B + I_C$$

And I_B is significantly smaller than I_C

Advantages:

CC amplifiers offer high current gain and low input impedance, making them suitable for applications where a high-resistance input needs to drive a low-resistance output load.

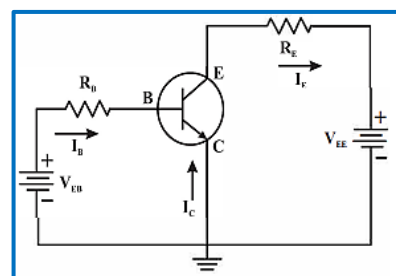
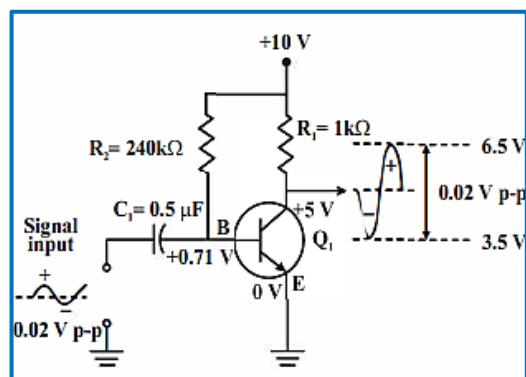
Applications:

Commonly used in impedance matching and signal buffering circuits due to their favorable characteristics.

1. COMMON BASE (CB) CIRCUIT:

In CB configuration, the input and output signals share the base terminal of the transistor, figure demonstrates this configuration.

Here, the input signal is applied to the emitter, and the output signal is taken from the collector.



The base-emitter junction is forward biased to allow transistor operation.

Characteristics:

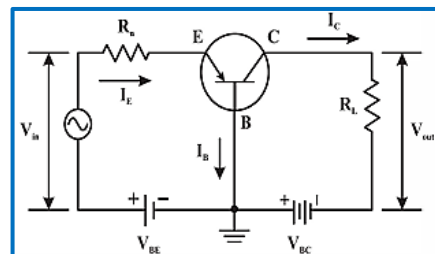
CB amplifiers provide low input impedance and approximately unity voltage gain.

The current gain in CB configuration is expressed as $\frac{I_{out}}{I_{in}}$ or the formula $\frac{I_C}{I_E}$

However, as the base current I_B is extremely small compared to the collector current I_C , the emitter current I_E is therefore approximately equal to the collector current. Thus $I_E \approx I_C$

Applications:

- Less commonly used compared to CC and Common Emitter (CE) configurations.
- CB amplifiers are employed in specific applications where low input impedance and unity voltage gain are advantageous.



OPERATIONAL AMPLIFIER:

An operational amplifier is a direct coupled high gain amplifier. It can be operated on both with AC and DC signals.

Definition:

Operational Amplifier is an integrated circuit, which can be used to perform various linear, non-linear, and mathematical operations.

Operational Amplifier, also called an Op-Amp.

Uses:

Op-amps are ideal for DC amplification and are used often in signal conditioning, filtering or other mathematical operations (add, subtract, integration and differentiation).

PROPERTIES OF AN IDEAL OPERATIONAL AMPLIFIER (OP-AMP):

The ideal op-amp has the following properties:

1. Infinite open loop voltage gain:

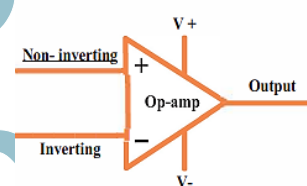
- An ideal op amp is a device often used as an amplifier. When you input voltage into it, the op amp outputs an amplified voltage.
- In an ideal scenario, the op amp would provide extremely high gain, essentially infinite, amplifying the signal countless times for maximum gain as needed.

2. Infinite input resistance (Impedance)

- An ideal op amp has super high input impedance, which means it won't load the circuit. If input impedance is low, the op amp draws more current; if it's high, less current is drawn.
- We aim for high input impedance to avoid disturbing the original circuit by minimizing current pulled from it, ideally with infinite Input impedance.

3. Zero output resistance (Impedance)

- In an ideal op amp, the output impedance is zero. This means when the op amp generates a signal, we want it to have zero resistance, ensuring that the maximum voltage goes to the output load.
- In a circuit, voltage gets divided based on the impedance. Higher impedance means more voltage drop. For the voltage to drop across the output load, the load's impedance should be greater than the op amp's output. That's why, ideally, we aim for zero output impedance in the op amp.



4. Gain independent of frequency

In an ideal op amp, the produced gain remains constant regardless of the input signal's frequency. This means the amplification stays reliable and consistent across all frequencies.

5. Infinite bandwidth

The bandwidth of an op-amp is the range of frequencies it amplifies equally. An ideal op-amp amplifies all frequencies, so it should ideally have an infinite bandwidth.

6. Infinite slew rate

- An ideal op-amp should change the output instantaneously as the input is changed.
- The slew rate of the op-amp is the factor that affects this time delay. An infinite slew rate means there is no time delay.

7. Zero noise contribution

Any signal includes a small amount of noise. The ideal op-amp does not produce any noise itself, although it will amplify any noise that is present in its input.

OP-AMP AS COMPARATOR:

The op-amp shown in figure is connected to two power supplies.

1. One battery of 9V is connected between the zero-volt line and the +9V positive supply terminal of the op-amp and
2. The other is between the zero-volt line and the negative power supply terminal. These batteries are not shown.

The output voltage is given by:

$$V_{out} = G_o \times (V^+ - V^-)$$

Where G_o is the open-loop voltage gain.

All of these voltages are measured with reference to the zero-volt line, which is often connected to earth.

Example:

Suppose that: $G_o = 10^5$, $V^+ = 0.15 \text{ V}$, $V^- = 0.10 \text{ V}$.

Then equation becomes:

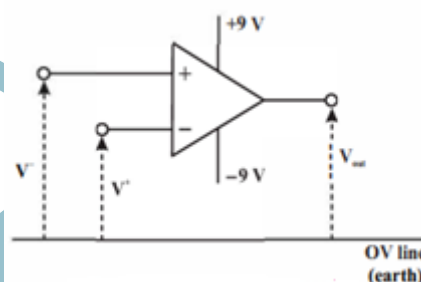
$$V_{out} = 10^5 \times (0.15 - 0.10) = 5000 \text{ V}$$

Certainly, it's impossible. The op-amp is therefore saturated and V_{out} will be close to one of the power supply voltages (in this case 9V) because V^+ is bigger than V^- .

In fact:

- If V^+ is slightly greater in magnitude than V^- , then V_{out} will have magnitude equal to the positive power supply voltage.
- If V^+ is slightly smaller in magnitude than V^- , then V_{out} will have magnitude equal to the negative power supply voltage.

The op-amp is serving as a comparator, comparing V^+ and V^- . If these voltages are slightly different, then the output voltages tell us which one is larger.



EFFECTS OF NEGATIVE FEEDBACK ON THE GAIN OF AN OPERATIONAL AMPLIFIER:

NEGATIVE FEEDBACK:

Negative feedback occurs when a portion of the output signal, like voltage or current, is feedback into the input in a way that opposes or subtracts from the input signal.

Advantages:

1. Reduced Gain: Negative feedback decreases the overall gain of the system.

2. Decreased Distortion and Noise: It minimizes distortion and noise, leading to a cleaner output signal.
3. Improved Stability: The gain becomes more stable and less sensitive to external changes, such as temperature.
4. Increased Bandwidth: Enhances the system's bandwidth.
5. Better Impedance: Lowers the output impedance and raises the input impedance.

Operation in Electronic Systems:

Feedback is unidirectional, flowing from output to input, ensuring the loop gain (G) remains independent of load and source impedances. A summing point at the input subtracts the feedback signal from the input signal, creating an error signal that drives the system. Example Circuit (Figure 2.28):

Setup:

Input V_{in} is connected to the non-inverting input (+) of an op-amp. The output V_{out} is fed back to the inverting input (-).

Operation:

With an infinite open-loop gain and no saturation, V^- equals V^+ . As, V_{in} changes, V_{out} adjusts to maintain, $V^- = V^+$

Equation:
$$V_{out} = G_o \times (V^+ - V^-)$$

Simplified: Since $V_{out} = V^-$ and $V_{in} = V^+$ then we have

$$V_{out} = G_o (V_{in} - V_{out})$$

$$V_{out} (1 + G_o) = G_o V_{in}$$

The closed loop gain G is given by:
$$G = \frac{V_{out}}{V_{in}} = \frac{G_o}{1 + G_o} \dots\dots\dots 22.5$$

As G_o is very high about 10^5 , there is little difference between G_o and $(1 + G_o)$, so the closed-loop gain is very nearly 1. This analysis is true as long as the output voltage is smaller than the supply voltage, in this case as long as V_{out} is between -6 v and +6 v.

Practical Example:

A Piezo-electric microphone with high internal resistance can be connected to an op-amp to increase current output while maintaining the same voltage.

Negative Feedback Benefits Summary:

- Less distortion.
- Increased bandwidth.
- The gain is more stable and not affected by changes in temperature, etc.

The output resistance (impedance) can be low and the input resistance high.

CIRCUIT DIAGRAMS FOR SINGLE SIGNAL INPUT:

INVERTING OPERATIONAL AMPLIFIER:

In inverting operational amplifiers, the op amp forces the negative terminal to equal the positive terminal, which is commonly ground. Therefore, the input-current is determined by the V_{in}/R_1 ratio as shown in Figure.

In this configuration, the same current flows through R_2 to the output. Ideally, current does not flow into the operational amplifier's negative terminal due to its high Z_{in} . The current flowing from the negative terminal through R_2 creates an inverted voltage polarity with respect to V_{in} . This is why these op amps are labeled with an inverting configuration. Note that the op amp's output can only swing between its positive and negative supplies, so creating a negative output voltage requires an op amp with a negative supply rail. V_{out} can be calculated with Equation

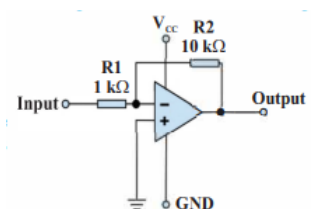


Figure 22.29
circuit diagrams for inverting Amp

$$V_{out} = \frac{R_2}{R_1} \times V_{in} \dots\dots\dots 22.6$$

NON-INVERTING OPERATIONAL AMPLIFIER:

In a non-inverting amplifier circuit, input signal from the source is connected to the non-inverting (+) terminal as shown in figure 22.30. Non-Inverting Operational Amplifier.

The operational amplifier forces the inverting (-) terminal voltage to equal the input voltage, which creates a current flow through the feedback resistors. The output voltage is always in phase with the input voltage, which is why this topology is known as non-inverting. Note that with a non-inverting amplifier, the voltage gain is always greater than 1, which is not always the case with the inverting configurations. V_{out} can be calculated with Equation:

$$V_{out} = \frac{1+R_2}{R_1} \times V_{in} \dots\dots\dots 22.6$$

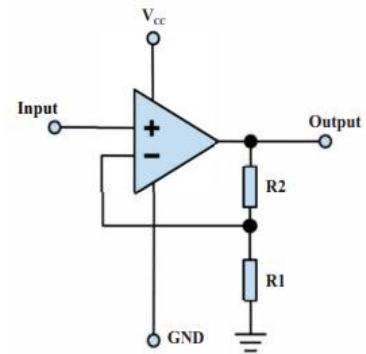


Figure 22.30 circuits for non-inverting amplifier

VIRTUAL EARTH APPROXIMATION:

To understand how the inverting amplifier works, you need to understand the concept of virtual earth approximation. In this approximation the potential at the inverting input (-) is very close to 0 V. There are two steps in the arguments:

The op-amps multiplies the difference in potential between the inverting and non-inverting inputs, V^- and V^+ to produce the output voltage V_{out} .

Because the open-loop voltage gain is very high, the difference between V^- and V^+ must be almost zero.

The non-inverting input (+) is connected to the zero-volt line so $V^+ = 0$. Thus, V^- must be close to zero and the inverting input (-) is almost at earth potential.

Point A is known as a virtual earth as shown in figure 22.31. It cannot actually be 0V, but it is very close to 0V. This approximation is true as long as the op-amp is not saturated, and for frequencies where the open-loop voltage gains is high.

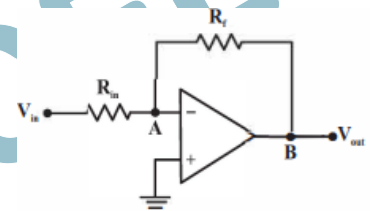


Figure 22.31 virtual earth approximation

Applications:

This approximation is often used in summing amplifier circuits and other applications where it simplifies analysis.

Summing Amplifier Example:

- In a summing amplifier, multiple input voltages can be summed using resistors connected to the inverting input.
- The virtual ground assumption makes it easier to analyze the currents and voltages in the circuit.

GAIN OF AN INVERTING AMPLIFIER:

If the current in the input resistor R_{in} is I_{in} and the current in the feedback resistor R_f is I_f , then point P is at 0V:

i.e.
$$I_{in} = \frac{R_{in}}{R_{in}} \text{ and } I_f = \frac{V_{out}}{R_f}$$

The input resistance of the op-amp is very high and so virtually no current enters or leaves the inverting input (-) of the op-amp. This means that I_{in} and I_f must be equal in size.

If V_{in} is positive potential, then the current in two resistors flows from left to right.

V_{out} will be negative because the current flows from P, which is at 0V, to the output connection, which must have a lower voltage than 0V.

Thus
$$I_f = -I_{in} \text{ and } \frac{V_{out}}{R_f} = -\frac{R_{in}}{R_{in}}$$

The gain of the inverting-amplifier is thus given by
$$G = \frac{V_{out}}{R_{in}} = -\frac{R_f}{R_{in}}$$

The negative sign shows that when the input voltage is positive the output voltage is negative and when the input is positive the output is negative. If the input voltage is alternating, then there will be a phase difference of 180° or π rad between the input and the output voltages.

SUMMARY

- A p-type substance is a semiconductor material which has an excess of positively charged particles called "holes" as its majority charge carriers.
- An n-type substance is a semiconductor material that has an excess of negatively charged electrons as its majority charge carriers.
- An intrinsic semiconductor is pure semiconductor material with no intentional impurities added.
- Doping is the process of intentionally introducing impurities into a semiconductor crystal to modify its electrical properties.
- Doping with Donor Impurities: A small amount of a group V element is introduced into the crystal lattice of the semiconductor. Common donor impurities include phosphorus (P) or arsenic (As).
- Doping with Acceptor Impurities: A small amount of a group III element is introduced into the crystal lattice of the semiconductor. Common acceptor impurities include boron (B) or gallium (Ga).
- P-N Junction is formed when the p-type and n-type semiconductors are joined, is called as p-n junction.
- P-N Junction Diode is a semiconductor device that is formed through this method and used in allowing the flow of electric current in one direction and blocking in the opposite.
- Forward Bias: The $P - N$ junction is forward-biased when the $P - type$ is connected to the positive terminal of the battery and the $N - type$ is connected to the negative terminal.
- Reverse Bias: The $P - N$ junction is reverse biased when the $P - type$ is connected to the negative terminal of the battery and the $N - type$ is connected to the positive side.
- Rectification is the process of converting an alternating current (AC) waveform into a direct current (DC) waveform, i.e., creating a new waveform that has only a single polarity.
- A light -emitting diode (LED) is a $P - N$ junction in which current flows as it becomes forward biased.
- A photodiode is one type of light detector which is used to convert the light into current or voltage based on the mode of operation of the device.
- Photo Voltaic Cells or Solar cells, also called photovoltaic cells, are rather heavily doped p-n junction diodes used to convert sunlight into electric energy.
- Transistor: The Bipolar Junction Transistor was invented in 1948 by J. Bardeen, W. Shockley and W. Brattain.
- It consists of a crystal of one type of doped semiconductor sandwiched between two of the opposite type.

TEXTBOOK WORK

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. A semiconductor is an element with a valence electron _____
(a) Four (b) Eight (c) Two (d) One
2. A pure semiconductor is known as
(a) Extrinsic (b) Intrinsic (c) Transistor (d) Diode
3. An acceptor atom is also called
(a) Penta-valent atom (b) Trivalent atom (c) Minority carrier (d) Majority carrier
4. Which one of the following elements silicon should be doped so as to give p-type of semiconductor?
(a) Germanium (b) Arsenic (c) Selenium (d) Boron

5. For a full-wave rectifier, the output frequency (MP:25)
(a) Equals one-half the input frequency (b) Equals the line frequency
(c) Equals two times the input (d) Is three times the line frequency
6. LED construction needs a semiconductor material is: (2024)
(a) Silicon (b) Germanium (c) Gallium (d) Gallium arsenide
7. The frequency of a half-wave signal is
(a) Twice the line frequency (b) Equal to the line frequency
(c) One-half of the line frequency (d) One-fourth the line frequency
8. The voltage gain of an emitter follower circuit is
(a) High (b) Low (c) Very high (d) Moderate
9. What is also called the conventional amplifier?
(a) Common-collector circuit (b) Emitter follower circuit
(c) Common base circuit (d) Common emitter circuit
10. An op-amp with negative feedback provides __ output parameter.
(a) Gain (b) Bandwidth
(c) Input-output impedance (d) Amplified

SECTION- B: SHORT ANSWERED QUESTIONS

1. Describe a p-n junction (diode) and how holes and electrons are produced in semiconductor?
2. Define and distinguish between p-n-p & n-p-n transistors?
3. Explain common-base and common collector configurations?
4. Describe the operations of transistors.
5. Explain the use of transistors as a switch and an amplifier (common emitter).
6. How would you understand the effects of negative feedback on the gain of an operational amplifier?
7. Draw and briefly describe the circuit diagrams for both the inverting and the noninverting amplifier for single signal input?
8. Derive an expression for the gain of inverting amplifiers by using virtual earth approximation.
9. Describe the properties of an ideal operational amplifier

SECTION- C: LONG ANSWERED QUESTIONS

1. Define Intrinsic (pure) and doped semiconductors. And How the N-type and P-type semiconductors are produced
2. Describe a P-N junction (diode) discuss its forward and reverse biasing. (2024)
3. Describe the I-V characteristic curves of P-N junction.
4. Define rectification and describe the use of diodes for half and full wave rectifications.
5. Describe the function and use of LED, Photodiode and Photo voltaic cell Describe the function and use of LED, Photodiode and Photo voltaic cell.
6. Explain the use of transistors as a switch and an amplifier (common emitter).
7. Describe the properties of an ideal operational amplifier.

SELF ASSESSMENT

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. To obtain p-type semiconductor silicon must be doped with the atom whose valency is: **(2023)**
(a) 2 (b) 3 (c) 4 (d) 5
2. A semiconductor is an element with valence electrons:
(a) Four (b) Eight (c) Two (d) One
3. A pure semiconductor is known as
(a) Extrinsic **(b) Intrinsic** (c) Transistor (d) Diode
4. An acceptor atom is also called
(a) Penta-valent atom **(b) Trivalent atom** (c) Minority carrier (d) Majority carrier
5. With which one of the following elements silicon should be doped so as to give p-type of semiconductor?
(a) Germanium (b) Arsenic (c) Selenium **(d) Boron**
6. For a full wave rectifier, the output frequency:
(a) Equals one-half the input frequency (b) Equals the input frequency
(c) Equals two times the input frequency (d) Equals three times the input frequency **(MP:25)**

SECTION- B: SHORT ANSWERED QUESTIONS

7. Define valance band and conduction band, classify solids on the basis of energy bands. **(MP:25)**
8. In a common base connection $I_c = 0.95\text{mA}$ and $I_b = 0.05\text{mA}$. Find the value of α . **(2024)**
9. What is meant by Rectification? How does semiconductor diode work as a half wave rectifier? **(2023)**

SECTION- B: SHORT ANSWERED QUESTIONS

10. Describe a P-N junction (diode) discuss its forward and reverse biasing. **(2024)**